KCE PATHWAYS · GRADES 4–5 · UNIT 5: CHARTING THE COURSE · WEEK 24 · MATH

The Bridge Design Studio — Teacher Lesson Plan & Answer Key

Learning objectives: Students classify triangles by their sides (equilateral, isosceles, scalene) and by their angles (acute, right, obtuse), use the quadrilateral family tree to name trapezoids, parallelograms, rectangles, rhombuses and squares and explain that a shape keeps every feature of the shapes above it, find parallel and perpendicular sides inside figures, count lines of symmetry, and solve multistep perimeter and design problems.

Duration: 35–45 minutes
Materials: Computer or tablet with the lesson open · pencil · printed Pre & Post sheet

Lesson flow (as it appears on screen)

1. Slide 0 — Intro and name: meet Ximena (10) in Pittsburgh, Pennsylvania, where her Aunt Winnie is a structural engineer at a bridge design studio. Ms. Whitaker, the lead engineer, checks every drawing, Mr. Wojcik cuts steel plates for the scale models, and Ximena's little brother Xander and her friend Wren drop by. The student types first and last name so results are sent. Read the scoring rules together: 2 tries per question, Pathway Points only for a correct first try, and the answer plus the reason appear after the 2nd miss. Typed number answers ignore capital letters, commas in large numbers, dollar signs and a final period.
2. Slide 1 — Warm-Up, Part 1: Line Plots with Fractions Review (4 pts, 2 each, Week 23 skill). If a student misses both warm-up slides, rename one pair of fractions with a common denominator before moving on.
3. Slide 2 — Warm-Up, Part 2 (4 pts, 2 each): a difference with renaming and a fair share.
4. Slide 3 — Pencil Break (Pre): the student opens or prints the Pre/Post sheet (grade4-5/prepost/KCE_Math_PrePost_G4-5W24.pdf) and completes Page 1 (Pre-Lesson Warm-Up) off screen before any teaching.
5. Slide 4 — Why Bridge Designers Sort Shapes: nonfiction connection about Pittsburgh, where the Allegheny and Monongahela Rivers meet to form the Ohio River. The "City of Bridges" has 446 of them, including the Three Sisters (1920s), named for Roberto Clemente, Andy Warhol and Rachel Carson. Trusses such as the Smithfield Street Bridge use triangles, which keep their shape when pushed. No questions.
6. Slide 5 — Picture It: Sorting Triangles Two Ways. By sides: equilateral, isosceles (at least 2 equal sides), scalene. By angles: acute, right, obtuse. Every triangle gets two names, such as right isosceles. An equilateral triangle is a special isosceles triangle and is always acute; one obtuse angle makes an obtuse triangle.
7. Slide 6 — Your Turn, Part 1: Classifying Triangles (8 pts, 4 each): one name by sides or angles, and a full two-word name.
8. Slide 7 — Your Turn, Part 2 (4 pts): a perimeter problem with an isosceles, equilateral or scalene triangle.
9. Slide 8 — Teach: The Quadrilateral Family Tree. Quadrilateral → trapezoid (at least 1 pair of parallel sides, the definition KCE Pathways uses) and kite; trapezoid → parallelogram → rectangle and rhombus → square. Every shape keeps all the features of the shapes above it, so every square is a rectangle, but not every rectangle is a square.
10. Slide 9 — Your Turn, Part 1: The Family Tree (8 pts, 4 each): shape riddles and always/sometimes/never statements.

11. Slide 10 — Your Turn, Part 2 (4 pts): perimeter using the equal opposite sides of a parallelogram or rectangle.
12. Slide 11 — Teach: Parallel and Perpendicular Sides. Rectangles and squares: 2 pairs of parallel sides and 4 pairs of perpendicular sides. A parallelogram or rhombus with no right angles: 0 perpendicular sides. A right trapezoid: 1 pair of parallel sides and 2 right angles. Triangles never have parallel sides; a right triangle has 1 pair of perpendicular sides. A regular hexagon has 3 pairs of parallel sides.
13. Slide 12 — Your Turn, Part 1: Parallel and Perpendicular Sides (8 pts, 4 each).
14. Slide 13 — Your Turn, Part 2 (4 pts): name a right trapezoid, or add up right angles or perpendicular pairs across two shapes.
15. Slide 14 — Vocabulary Round-Up: 8 flip cards (equilateral triangle, isosceles triangle, scalene triangle, quadrilateral, parallelogram, rhombus, perpendicular sides, line of symmetry).
16. Slide 15 — Your Turn: Using the Words (6 pts, 3 each): one multiple-choice and one typed vocabulary item.
17. Slide 16 — Teach: Lines of Symmetry. Square 4; rectangle that is not a square 2 (its diagonals are NOT lines of symmetry); equilateral triangle 3; isosceles 1; scalene 0; rhombus 2 (along its diagonals); isosceles trapezoid 1; slanted parallelogram 0; regular hexagon 6.
18. Slide 17 — Your Turn: Lines of Symmetry (6 pts, 3 each): count the lines for one shape, and pick the shape with exactly the number asked.
19. Slide 18 — Real World: Measuring the Truss (4 pts): a bottom beam made of equilateral-triangle sides, an isosceles trapezoid panel's perimeter, or railing along three sides of a rectangular deck.
20. Slide 19 — Real World: Cutting Steel Plates (4 pts): the lines of symmetry of a rectangle with an isosceles triangle on top (1), and the full name of the pieces when a square or a rectangle is cut on a diagonal.
21. Slide 20 — Think It Through: Explain and Find the Mistake (10 pts, 5 each): a "which explanation is correct?" item and a "find the mistake" item. Ask the student to say why each wrong choice is wrong.
22. Slide 21 — Fix the Sentence, Part 1 (4 pts, 2 each): typed sentences checked exactly (capital letters, quotation marks, commas, end marks, and one grammar fix). Straight or curly quotes and extra spaces are OK. The corrected sentence appears after the 2nd miss.
23. Slide 22 — Fix the Sentence, Part 2 (4 pts, 2 each): two more sentences, same rules. Both serial-comma and no-serial-comma forms of a list are accepted where a list appears.
24. Slide 23 — Match-Up Round (10 pts): 5 clues (sides, angles, parallel sides or lines of symmetry) matched to the figures they describe, 2 pts each on the first try.
25. Slide 24 — Speed Trail (bonus XP only): 3 quick items against the student's own best time.
26. Slide 25 — Looking Back reflection (bonus XP only).
27. Slide 26 — Boss Battle: The Footbridge Design (8 pts, 4 each): a different situation in each set (3 isosceles trapezoid railing panels, a truss of equilateral triangles, or square plates cut along both diagonals).
28. Slide 27 — Pencil Break (Post): the student completes Page 2 (Post-Lesson Challenge) of the same sheet (grade4-5/prepost/KCE_Math_PrePost_G4-5W24.pdf), plus the ★ Challenge if time allows, before seeing the score.
29. Slide 28 — Results (Design Day Complete!): the score out of 100 is sent to KingCretot Experience. 85 or higher shows "Pittsburgh Bridge Design Master!" with the congratulations message "Your footbridge design passed every check in Pittsburgh!" and a big confetti burst; below 85 shows "Good Work at the Design Studio!" with encouragement ("You are getting closer every time."), and the next attempt brings a fresh question set (sets rotate 1, 2, 3). Both messages remind the student to finish the Post-Lesson Challenge sheet.

Lesson answer key with reasoning

Every scored item the student sees, in order, across all three rotating question sets. Each item allows 2 tries and earns Pathway Points only on a correct first try. The "why" is the same reasoning shown on screen after a second missed try.

WARM-UP: LINE PLOTS WITH FRACTIONS REVIEW (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
r1_s0	Set 1: Line plots (Week 23): a line plot of bolt lengths shows 3 X's above 3/8 inch and 2 X's above 1/2 inch. What is the total length of those 5 bolts? Inches:	**2 1/8** (also accepted, with or without the unit word: 2 2/16, 17/8, 2-1/8, 2 and 1/8, 2.125)	3 × 3/8 = 9/8. 2 × 1/2 = 1 = 8/8. 9/8 + 8/8 = 17/8 = 2 1/8 inches.
r2_s0	Set 1: Line plots (Week 23): where does the X for 4/8 inch go on a line plot? [In the same stack as 1/2 inch / In the same stack as 1/4 inch / Between 1/2 and 5/8 / Above the number 4]	**In the same stack as 1/2 inch**	4/8 and 1/2 are equivalent fractions, so they are the same point on the number line.
r3_s0	Set 1: Line plots (Week 23): on a line plot, the longest bolt is 1 5/8 inches and the shortest is 3/4 inch. What is the difference? Inches:	**7/8** (also accepted: 7/8 inches, 7/8 inch, 7/8 in, 14/16, 14/16 inches, 14/16 inch, 14/16 in, 0.875, 0.875 inches, 0.875 inch, 0.875 in)	Rename: 1 5/8 = 13/8 and 3/4 = 6/8. 13/8 − 6/8 = 7/8 inch.
r4_s0	Set 1: Fair share (Week 23): four cups hold 1/4, 1/2, 3/4, and 1/2 cup of paint. If the paint were shared equally, how much would each cup hold? Cups:	**1/2** (also accepted, with or without the unit word: 2/4, 4/8, 8/16, 0.5)	Total: 1/4 + 1/2 + 3/4 + 1/2 = 2 cups. Share among 4: 2 ÷ 4 = 2/4 = 1/2 cup.
r1_s1	Set 2: Line plots (Week 23): a line plot of washer widths shows 4 X's above 5/8 inch and 1 X above 3/4 inch. What is the total width of those 5 washers? Inches:	**3 1/4** (also accepted, with or without the unit word: 3 2/8, 3 4/16, 13/4, 3-1/4, 3 and 1/4, 3.25)	4 × 5/8 = 20/8. 3/4 = 6/8. 20/8 + 6/8 = 26/8 = 3 2/8 = 3 1/4 inches.
r2_s1	Set 2: Line plots (Week 23): the data are 1/4, 3/8, 1/2, and 3/4 inch. Which tick marks does the number line need? [Eighths / Halves / Fourths / Wholes]	**Eighths**	3/8 is in eighths, so the number line needs eighths. Eighths also land on 1/4, 1/2, and 3/4.
r3_s1	Set 2: Line plots (Week 23): a cable is 2 inches long and a wire is 3/8 inch long. How much longer is the cable? Inches:	**1 5/8** (also accepted, with or without the unit word: 1 10/16, 13/8, 1-5/8, 1 and 5/8, 1.625)	Rename 2 as 1 8/8. 1 8/8 − 3/8 = 1 5/8 inches.
r4_s1	Set 2: Fair share (Week 23): five cups hold 1/4, 1/2, 3/4, 1, and 1 1/4 cups of paint. If the paint were shared equally, how much would each cup hold? Cups:	**3/4** (also accepted, with or without the unit word: 6/8, 12/16, 0.75)	Total in fourths: 1 + 2 + 3 + 4 + 5 = 15/4 cups. Share among 5: 15/4 ÷ 5 = 3/4 cup.
r1_s2	Set 3: Line plots (Week 23): a line plot of rivet lengths shows 2 X's above 1 1/4 inches and 2 X's above 7/8 inch. What is the total length of those 4 rivets? Inches:	**4 1/4** (also accepted, with or without the unit word: 4 2/8, 4 4/16, 17/4, 4-1/4, 4 and 1/4, 4.25)	2 × 1 1/4 = 2 2/4 = 2 1/2. 2 × 7/8 = 14/8 = 1 3/4. 2 1/2 + 1 3/4 = 2 2/4 + 1 3/4 = 3 5/4 = 4 1/4 inches.
r2_s2	Set 3: Fair share (Week 23): 4 jars hold a total of 3 cups of water. If the water were shared equally, how much would each jar hold? [3/4 cup / 1 1/3 cups / 3 cups / 1/4 cup]	**3/4 cup**	Fair share = total ÷ number of jars = 3 ÷ 4 = 3/4 cup.
r3_s2	Set 3: Line plots (Week 23): the longest washer on a line plot is 1 1/4 inches and the shortest is 5/8 inch. What is the difference? Inches:	**5/8** (also accepted: 5/8 inches, 5/8 inch, 5/8 in, 10/16, 10/16 inches, 10/16 inch, 10/16 in, 0.625, 0.625 inches, 0.625 inch, 0.625 in)	Rename 1 1/4 as 10/8. 10/8 − 5/8 = 5/8 inch.

ID	Question	Answer	Why (explain to student)
r4_s2	Set 3: Fair share (Week 23): four cups hold 3/8, 5/8, 1, and 1 1/2 cups of paint. If the paint were shared equally, how much would each cup hold? Cups:	**7/8** (also accepted: 7/8 cups, 7/8 cup, 7/8 c, 14/16, 14/16 cups, 14/16 cup, 14/16 c, 0.875, 0.875 cups, 0.875 cup, 0.875 c)	Total in eighths: 3 + 5 + 8 + 12 = 28/8 = 3 1/2 cups. Share among 4: 28/8 ÷ 4 = 7/8 cup.

YOUR TURN: CLASSIFYING TRIANGLES (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
l1_s0	Set 1: Aunt Winnie's truss drawing has a triangle with sides of 9 feet, 9 feet, and 9 feet. Classify it by its sides. (one word)	**equilateral** (also accepted: equilateral triangle; with "a" or "an" or in the plural)	All three sides are equal, so the triangle is equilateral.
l2_s0	Set 1: A gusset plate is a triangle with angles of 90°, 45°, and 45°, and two sides that are each 6 inches long. What is its full name? [Right isosceles triangle / Acute isosceles triangle / Right scalene triangle / Obtuse equilateral triangle]	**Right isosceles triangle**	It has one 90° angle, so it is right. It has two equal sides, so it is isosceles.
l3_s0	Set 1: Two steps: an isosceles triangle brace has a perimeter of 26 feet. Its third side, the one that is not equal to the others, is 8 feet. How long is each of the two equal sides? Feet:	**9** (also accepted: 9 feet, 9 foot, 9 ft)	The two equal sides together: 26 − 8 = 18 feet. Each one: 18 ÷ 2 = 9 feet. Check: 9 + 9 + 8 = 26.
l1_s1	Set 2: Aunt Winnie's truss drawing has a triangle with sides of 5 feet, 7 feet, and 10 feet. Classify it by its sides. (one word)	**scalene** (also accepted: scalene triangle; with "a" or "an" or in the plural)	No two sides are the same length, so the triangle is scalene.
l2_s1	Set 2: A roof brace is a triangle with angles of 30°, 30°, and 120°, and two equal sides. What is its full name? [Obtuse isosceles triangle / Acute isosceles triangle / Obtuse scalene triangle / Right isosceles triangle]	**Obtuse isosceles triangle**	The 120° angle is more than 90°, so it is obtuse. It has two equal sides, so it is isosceles. Two acute angles do not make it acute: one obtuse angle decides.
l3_s1	Set 2: Two steps: Mr. Wojcik builds 5 separate equilateral triangles for a model truss. Each side is 3 feet long. How many feet of steel bar does he use in all? Feet:	**45** (also accepted: 45 feet, 45 foot, 45 ft)	One equilateral triangle: 3 + 3 + 3 = 9 feet. Five triangles: 5 × 9 = 45 feet.
l1_s2	Set 3: Aunt Winnie's truss drawing has a triangle with angles of 50°, 60°, and 70°. Classify it by its angles. (one word)	**acute** (also accepted: acute triangle; with "a" or "an" or in the plural)	All three angles are less than 90°, so the triangle is acute.
l2_s2	Set 3: A support plate is a triangle with angles of 90°, 35°, and 55°, and no equal sides. What is its full name? [Right scalene triangle / Right isosceles triangle / Acute scalene triangle / Obtuse scalene triangle]	**Right scalene triangle**	It has one 90° angle, so it is right. No sides are equal, so it is scalene.
l3_s2	Set 3: Two steps: a scalene triangle panel has a perimeter of 30 feet. One side is 8 feet and another is 12 feet. How long is the third side? Feet:	**10** (also accepted: 10 feet, 10 foot, 10 ft)	The two known sides: 8 + 12 = 20 feet. Third side: 30 − 20 = 10 feet. All three sides are different, so it really is scalene.

YOUR TURN: THE FAMILY TREE (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
k1_s0	Set 1: Riddle: I have 4 sides, 2 pairs of parallel sides, and 4 right angles, but my sides are not all the same length. What am I? (one word)	**rectangle** (also accepted with “a” or “an” or in the plural)	Two pairs of parallel sides make a parallelogram. Four right angles make it a rectangle. The sides are not all equal, so it is not a square.
k2_s0	Set 1: Which statement is ALWAYS true? [Every square is a rectangle / Every rectangle is a square / Every parallelogram is a rectangle / Every rhombus is a square]	**Every square is a rectangle**	A square has 4 right angles, so it is always a rectangle. The other shapes only sometimes have the extra features they would need.
k3_s0	Set 1: Two steps: a parallelogram-shaped deck panel has one side that is 9 feet long and a side next to it that is 5 feet long. What is its perimeter? Feet:	**28** (also accepted: 28 feet, 28 foot, 28 ft)	Opposite sides are equal, so the sides are 9, 5, 9, and 5. 2 × 9 + 2 × 5 = 18 + 10 = 28 feet.
k1_s1	Set 2: Riddle: I have 4 equal sides and 2 pairs of parallel sides, but I have no right angles. What am I? (one word)	**rhombus** (also accepted: rhombi; with “a” or “an” or in the plural)	A parallelogram with 4 equal sides is a rhombus. With no right angles, it cannot be a square.
k2_s1	Set 2: A square is ALWAYS which of these? [A rectangle and a rhombus / A rectangle, but never a rhombus / A rhombus, but never a rectangle / Neither a rectangle nor a rhombus]	**A rectangle and a rhombus**	A square has 4 right angles (so it is a rectangle) and 4 equal sides (so it is a rhombus).
k3_s1	Set 2: Two steps: a rectangle-shaped sign has a perimeter of 40 meters. One side is 12 meters long. How long is a side next to it? Meters:	**8** (also accepted: 8 meters, 8 meter, 8 m)	The two 12-meter sides use 2 × 12 = 24 meters. The other two sides share 40 − 24 = 16 meters, so each is 16 ÷ 2 = 8 meters.
k1_s2	Set 3: Riddle: I have 4 sides and exactly 1 pair of parallel sides. What am I? (one word)	**trapezoid** (also accepted with “a” or “an” or in the plural)	A quadrilateral with at least one pair of parallel sides is a trapezoid. With only one pair, it cannot be a parallelogram.
k2_s2	Set 3: Always, sometimes, or never: a rhombus is a square. [Sometimes, only when it has 4 right angles / Always / Never / Sometimes, only when it has 1 pair of parallel sides]	**Sometimes, only when it has 4 right angles**	Every rhombus has 4 equal sides. It is a square only if its angles are all right angles too.
k3_s2	Set 3: Two steps: a parallelogram-shaped steel plate has a perimeter of 34 feet. One side is 11 feet long. How long is a side next to it? Feet:	**6** (also accepted: 6 feet, 6 foot, 6 ft)	The two 11-foot sides use 2 × 11 = 22 feet. The other two sides share 34 − 22 = 12 feet, so each is 12 ÷ 2 = 6 feet.

YOUR TURN: PARALLEL AND PERPENDICULAR SIDES (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
e1_s0	Set 1: A square window in the bridge tower has how many pairs of perpendicular sides? Pairs:	**4** (also accepted: 4 pairs, 4 pair)	A square has 4 right angles. At each corner, two sides meet at 90°, so there are 4 pairs of perpendicular sides.
e2_s0	Set 1: Which shape can have EXACTLY two right angles? [A trapezoid / A rectangle / A triangle / A parallelogram that is not a rectangle]	**A trapezoid**	A right trapezoid has exactly 2 right angles. A rectangle always has 4, a triangle can have at most 1, and a parallelogram that is not a rectangle has none.

ID	Question	Answer	Why (explain to student)
e3_s0	Set 1: Ximena's sketch shows a four-sided panel with exactly 1 pair of parallel sides and exactly 2 right angles. What is the best name for the shape? (one or two words)	**right trapezoid** (also accepted with "a" or "an" or in the plural)	One pair of parallel sides makes it a trapezoid. Two right angles make it a right trapezoid.
e1_s1	Set 2: How many pairs of parallel sides does a triangle have? Pairs:	**0** (also accepted: 0 pairs, 0 pair, none, zero, no pairs)	Any two sides of a triangle meet at a corner, so they can never be parallel. The answer is 0.
e2_s1	Set 2: Which shape has 2 pairs of parallel sides but NO right angles? [A rhombus that is not a square / A rectangle / A square / A right trapezoid]	**A rhombus that is not a square**	A rhombus is a parallelogram, so it has 2 pairs of parallel sides. If it is not a square, none of its angles are right angles.
e3_s1	Set 2: Two steps: how many pairs of perpendicular sides does a right trapezoid have, plus the number in a right triangle? Pairs:	**3** (also accepted: 3 pairs, 3 pair)	A right trapezoid has 2 right angles, so 2 pairs of perpendicular sides. A right triangle has 1. 2 + 1 = 3 pairs.
e1_s2	Set 3: A bolt head is shaped like a regular hexagon. How many pairs of parallel sides does it have? Pairs:	**3** (also accepted: 3 pairs, 3 pair)	A regular hexagon has 6 sides, and each side is parallel to the side across from it. 6 sides make 3 pairs.
e2_s2	Set 3: Which triangle has a pair of perpendicular sides? [A right triangle / An acute triangle / An obtuse triangle / An equilateral triangle]	**A right triangle**	Perpendicular sides meet at a right angle, and only a right triangle has a right angle.
e3_s2	Set 3: Two steps: Mr. Wojcik cuts 2 square plates and 1 right triangle plate. How many right angles do the 3 plates have in all? Right angles:	**9** (also accepted: 9 right angles, 9 angles)	Each square has 4 right angles: 2 × 4 = 8. The right triangle has 1. 8 + 1 = 9 right angles.

YOUR TURN: USING THE WORDS (3 POINTS EACH)

ID	Question	Answer	Why (explain to student)
v1_s0	Set 1: From the lesson: "An isosceles triangle has at least two equal sides, so an equilateral triangle is a special isosceles triangle." Which triangle is isosceles? [Sides of 7, 7, and 4 inches / Sides of 5, 6, and 7 inches / Sides of 3, 8, and 9 inches / Sides of 2, 4, and 5 inches]	**Sides of 7, 7, and 4 inches**	7, 7, and 4 has two equal sides, so it is isosceles. The others have no equal sides, so they are scalene.
v2_s0	Set 1: Type your answer: "A ___ is any closed figure with four straight sides." (one word)	**quadrilateral** (also accepted with "a" or "an" or in the plural)	Quad means four and lateral means side: a quadrilateral has four sides.
v1_s1	Set 2: From the lesson: "A line of symmetry splits a figure into two halves that match exactly when you fold along the line." Which fold is a line of symmetry for a rectangle that is not a square? [A fold straight across the middle, from the midpoint of one side to the midpoint of the opposite side / A fold along a diagonal, from corner to corner / A fold along one of its sides / Any fold that goes through a corner]	**A fold straight across the middle, from the midpoint of one side to the midpoint of the opposite side**	Folding a rectangle across the middle makes the two halves match. Folding a non-square rectangle on a diagonal leaves the corners sticking out.
v2_s1	Set 2: Type your answer: "A ___ is a parallelogram with four equal sides." (one word)	**rhombus** (also accepted: rhombi; with "a" or "an" or in the plural)	A rhombus has four equal sides and two pairs of parallel sides.

ID	Question	Answer	Why (explain to student)
v1_s2	Set 3: From the lesson: "Two sides that meet at a right angle are perpendicular sides." Which shape has NO perpendicular sides? [A rhombus with no right angles / A rectangle / A right triangle / A right trapezoid]	**A rhombus with no right angles**	Perpendicular sides need a right angle. A rhombus with no right angles has none. The other three shapes each have at least one right angle.
v2_s2	Set 3: Type your answer: a triangle with no equal sides is a ___ triangle. (one word)	**scalene** (also accepted with "a" or "an" or in the plural)	Scalene means no two sides are the same length.

YOUR TURN: LINES OF SYMMETRY (3 POINTS EACH)

ID	Question	Answer	Why (explain to student)
t1_s0	Set 1: A square steel plate has how many lines of symmetry? Lines:	**4** (also accepted: 4 lines, 4 line)	A square matches when folded up and down, side to side, and along both diagonals: 4 lines of symmetry.
t2_s0	Set 1: Which shape has EXACTLY one line of symmetry? [An isosceles triangle with only two equal sides / A rectangle that is not a square / A scalene triangle / An equilateral triangle]	**An isosceles triangle with only two equal sides**	An isosceles triangle with only two equal sides folds onto itself only along the line down its middle. A rectangle has 2, a scalene triangle has 0, and an equilateral triangle has 3.
t1_s1	Set 2: An equilateral triangle gusset plate has how many lines of symmetry? Lines:	**3** (also accepted: 3 lines, 3 line)	Each line goes from one corner to the middle of the side across from it. Three corners make 3 lines of symmetry.
t2_s1	Set 2: Which shape has NO lines of symmetry? [A parallelogram with no right angles and two different side lengths / A rectangle / An isosceles trapezoid / A rhombus]	**A parallelogram with no right angles and two different side lengths**	A slanted parallelogram with unequal sides never matches itself when folded. A rectangle and a rhombus each have 2, and an isosceles trapezoid has 1.
t1_s2	Set 3: A bolt head shaped like a regular hexagon has how many lines of symmetry? Lines:	**6** (also accepted: 6 lines, 6 line)	A regular hexagon has 6 equal sides and 6 lines of symmetry: 3 through opposite corners and 3 through the middles of opposite sides.
t2_s2	Set 3: Which shape has EXACTLY two lines of symmetry? [A rectangle that is not a square / A square / An isosceles trapezoid / A scalene triangle]	**A rectangle that is not a square**	A rectangle that is not a square folds evenly only across the middle both ways: 2 lines. A square has 4, an isosceles trapezoid has 1, and a scalene triangle has 0.

REAL WORLD: MEASURING THE TRUSS (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
w1_s0	Set 1: *The Situation Aunt Winnie's footbridge truss is a row of equilateral triangles. Each triangle has a perimeter of 18 feet. The bottom beam of the truss is made of 4 triangle sides in a row.* Two steps: how long is the bottom beam? Feet:	**24** (also accepted: 24 feet, 24 foot, 24 ft)	Each side of an equilateral triangle: 18 ÷ 3 = 6 feet. Four sides in a row: 4 × 6 = 24 feet.

ID	Question	Answer	Why (explain to student)
w1_s1	Set 2: *The Situation Ms. Whitaker draws a side panel shaped like a trapezoid with exactly one pair of parallel sides. The top is 30 meters, the bottom is 50 meters, and each slanted side is 25 meters.* Two steps: the two slanted sides are equal, so what kind of trapezoid is it, and what is its perimeter? [An isosceles trapezoid with a perimeter of 130 meters / An isosceles trapezoid with a perimeter of 105 meters / A right trapezoid with a perimeter of 130 meters / A parallelogram with a perimeter of 80 meters]	**An isosceles trapezoid with a perimeter of 130 meters**	Equal slanted sides make an isosceles trapezoid. Perimeter: 30 + 50 + 25 + 25 = 130 meters. (105 counts a slanted side only once.)
w1_s2	Set 3: *The Situation A lookout deck on the bridge is a rectangle 40 feet long and 12 feet wide. A railing goes along both long sides and one short side. The other short side is open to the walkway.* Two steps: how many feet of railing are needed? Feet:	**92** (also accepted: 92 feet, 92 foot, 92 ft)	Opposite sides of a rectangle are equal, so the long sides are 40 feet each. Railing: 40 + 40 + 12 = 92 feet. (The perimeter, 104 feet, would include the open side.)

REAL WORLD: CUTTING STEEL PLATES (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
w2_s0	Set 1: *The Situation The front of a bridge tower is drawn as a rectangle with an isosceles triangle on top. The base of the triangle sits exactly on the top side of the rectangle, and the two shapes are centered.* Two steps: how many lines of symmetry does the whole tower shape have? [1 / 0 / 2 / 3]	**1**	The rectangle has 2 lines of symmetry and the isosceles triangle has 1, but the WHOLE shape matches only when folded down the center, through the top of the triangle. A fold across the middle would put the triangle on top of nothing.
w2_s1	Set 2: *The Situation Mr. Wojcik cuts a square steel plate in half along one diagonal, from one corner to the opposite corner.* Two steps: give the full name of each piece, by its angles and by its sides. (two words, like “acute scalene”)	**right isosceles** (also accepted: right isosceles triangle, isosceles right, isosceles right triangle; with “a” or “an” or in the plural)	Each piece keeps one corner of the square, which is a right angle, so it is right. Its two short sides are sides of the square, which are equal, so it is isosceles.
w2_s2	Set 3: *The Situation Mr. Wojcik cuts a rectangle-shaped plate that is 8 inches long and 3 inches wide along one diagonal.* Two steps: what is the full name of each of the two pieces? [Right scalene triangles / Right isosceles triangles / Acute scalene triangles / Obtuse isosceles triangles]	**Right scalene triangles**	Each piece keeps one right angle from the rectangle, so it is right. Its sides are 8 inches, 3 inches, and the diagonal, which is longer than both, so no sides are equal: scalene.

THINK IT THROUGH: EXPLAIN AND FIND THE MISTAKE (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
x1_s0	Set 1: Which explanation is correct? Is every square a rhombus? [Yes, because a square has 4 equal sides and 2 pairs of parallel sides / No, because a rhombus can never have right angles / No, because a square is a rectangle instead / Yes, because every quadrilateral is a rhombus]	**Yes, because a square has 4 equal sides and 2 pairs of parallel sides**	A rhombus is a parallelogram with 4 equal sides. A square has both features, so it is a rhombus. A shape can belong to more than one group.
x2_s0	Set 1: Find the mistake: Wren says a triangle with angles of 100°, 40°, and 40° is an acute triangle, because two of its angles are acute. What should she fix? [One obtuse angle makes it an obtuse triangle / Nothing. It is an acute triangle / It is a right triangle, because 100° is close to 90° / It is an equilateral triangle, because two angles match]	**One obtuse angle makes it an obtuse triangle**	A triangle is named for its largest angle when that angle is right or obtuse. 100° is more than 90°, so this is an obtuse (isosceles) triangle.
x1_s1	Set 2: Which explanation is correct? Can a triangle have two right angles? [No. The two sides that meet the bottom side at right angles would be parallel, so they could never meet to close the triangle / Yes. A right isosceles triangle has two right angles / Yes, if all three sides are equal / No, because a triangle can never have any right angle]	**No. The two sides that meet the bottom side at right angles would be parallel, so they could never meet to close the triangle**	Two lines that are both perpendicular to the same line are parallel, and parallel lines never meet. So a triangle can have at most one right angle.
x2_s1	Set 2: Find the mistake: Xander says a rectangle that is 6 inches long and 2 inches wide has 4 lines of symmetry, including both diagonals. What should he fix? [It has only 2 lines of symmetry; folding on a diagonal does not make the halves match / Nothing. It has 4 lines of symmetry / It has 0 lines of symmetry / It has only 1 line of symmetry]	**It has only 2 lines of symmetry; folding on a diagonal does not make the halves match**	Fold a long, thin rectangle on a diagonal and the corners stick out past each other. Only the two folds across the middle work, so it has 2 lines of symmetry.
x1_s2	Set 3: Which explanation is correct? Is every rectangle a parallelogram? [Yes, because a rectangle has 2 pairs of parallel sides / No, because a parallelogram must have slanted sides / No, because a rectangle has right angles / Yes, because a rectangle has 4 equal sides]	**Yes, because a rectangle has 2 pairs of parallel sides**	A parallelogram is any quadrilateral with 2 pairs of parallel sides. A rectangle has them, so it is a parallelogram. Its sides do not have to be equal.
x2_s2	Set 3: Find the mistake: Wren says a trapezoid can never have a right angle. What should she fix? [A right trapezoid has 2 right angles, so a trapezoid can have right angles / Nothing. A trapezoid never has a right angle / A trapezoid always has 4 right angles / A trapezoid can have exactly 1 right angle, but never 2]	**A right trapezoid has 2 right angles, so a trapezoid can have right angles**	A right trapezoid has one side that meets both parallel sides at 90°, which makes 2 right angles. Right angles in a trapezoid come in twos.

FIX THE SENTENCE (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
s1_s0	Set 1: Fix this sentence: ximena asked, is a square also a rectangle	**Ximena asked, "Is a square also a rectangle?"**	Capitalize the name Ximena, put quotation marks around her exact words, capitalize the first word of the quote, and put the question mark inside the closing quotation mark.
s2_s0	Set 1: Fix this sentence: The bridge crosses the allegheny river in pittsburgh.	**The bridge crosses the Allegheny River in Pittsburgh.**	The names of rivers and cities are proper nouns, so capitalize Allegheny River and Pittsburgh.
s3_s0	Set 1: Fix this sentence: The truss uses triangles rectangles and trapezoids.	**The truss uses triangles, rectangles, and trapezoids.** (also accepted: The truss uses triangles, rectangles and trapezoids.)	Use commas to separate the items in a series: triangles, rectangles, and trapezoids.
s4_s0	Set 1: Fix this sentence: each of the triangles have three sides	**Each of the triangles has three sides.**	Capitalize the first word. The subject is each (one triangle at a time), so use has, not have. End with a period.
s1_s1	Set 2: Fix this sentence: ms. whitaker said, check every angle twice	**Ms. Whitaker said, "Check every angle twice."**	Capitalize Ms. and the name Whitaker, put quotation marks around her exact words, capitalize the first word of the quote, and put the period inside the closing quotation mark.
s2_s1	Set 2: Fix this sentence: The engineers checked they're drawings for mistakes.	**The engineers checked their drawings for mistakes.**	Their shows ownership: the drawings belong to the engineers. They're means they are.
s3_s1	Set 2: Fix this sentence: Mr. Wojcik a steel worker cut the plates.	**Mr. Wojcik, a steel worker, cut the plates.**	A steel worker describes Mr. Wojcik. Put a comma before and after a phrase like this that renames the person.
s4_s1	Set 2: Fix this sentence: the sides of a rhombus is all the same length	**The sides of a rhombus are all the same length.**	Capitalize the first word. The subject is sides (plural), not rhombus, so use are, not is. End with a period.
s1_s2	Set 3: Fix this sentence: xander said, that bridge looks like a giant triangle	**Xander said, "That bridge looks like a giant triangle."** (also accepted: Xander said, "That bridge looks like a giant triangle!")	Capitalize the name Xander, put quotation marks around his exact words, capitalize the first word of the quote, and put the end mark inside the closing quotation mark.
s2_s2	Set 3: Fix this sentence: A square is a rhombus to.	**A square is a rhombus too.**	Too means also: a square is a rhombus, also. To is used for direction, as in to the bridge.
s3_s2	Set 3: Fix this sentence: Yes the square is also a parallelogram.	**Yes, the square is also a parallelogram.**	Put a comma after an opening word like Yes, No, or Well.
s4_s2	Set 3: Fix this sentence: the beams on the side of the bridge is parallel	**The beams on the side of the bridge are parallel.**	Capitalize the first word. The subject is beams (plural), not side or bridge, so use are, not is. End with a period.

MATCH-UP ROUND (2 POINTS PER FIRST-TRY MATCH)

ID	Question	Answer	Why (explain to student)
match_s0_1	Set 1: Match 4 equal sides, 4 right angles	**square**	2 points if matched on the first try.
match_s0_2	Set 1: Match exactly 1 pair of parallel sides	**trapezoid**	2 points if matched on the first try.
match_s0_3	Set 1: Match 3 equal sides	**equilateral triangle**	2 points if matched on the first try.
match_s0_4	Set 1: Match one 90° angle	**right triangle**	2 points if matched on the first try.
match_s0_5	Set 1: Match 4 equal sides, no right angles	**rhombus**	2 points if matched on the first try.
match_s1_1	Set 2: Match 4 right angles, sides not all equal	**rectangle**	2 points if matched on the first try.
match_s1_2	Set 2: Match no equal sides	**scalene triangle**	2 points if matched on the first try.
match_s1_3	Set 2: Match 2 pairs of parallel sides, no right angles, sides not all equal	**parallelogram**	2 points if matched on the first try.
match_s1_4	Set 2: Match one angle greater than 90°	**obtuse triangle**	2 points if matched on the first try.
match_s1_5	Set 2: Match exactly 2 equal sides	**isosceles triangle**	2 points if matched on the first try.
match_s2_1	Set 3: Match 6 lines of symmetry	**regular hexagon**	2 points if matched on the first try.
match_s2_2	Set 3: Match 4 lines of symmetry	**square**	2 points if matched on the first try.
match_s2_3	Set 3: Match 3 lines of symmetry	**equilateral triangle**	2 points if matched on the first try.
match_s2_4	Set 3: Match exactly 2 lines of symmetry and 4 right angles	**rectangle**	2 points if matched on the first try.
match_s2_5	Set 3: Match 3 sides, no lines of symmetry	**scalene triangle**	2 points if matched on the first try.

BOSS BATTLE: THE FOOTBRIDGE DESIGN (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
b1_s0	Set 1: *Ximena and Aunt Winnie Finish the Design Ximena designs a railing panel shaped like an isosceles trapezoid with exactly one pair of parallel sides. The top is 10 feet, the bottom is 16 feet, and each slanted side is 5 feet. The footbridge needs 3 of these panels.* How many lines of symmetry does one panel have? [1 / 0 / 2 / 4]	**1**	An isosceles trapezoid matches only when folded down the middle, through the centers of the top and bottom. A fold across the middle would put the 10-foot top on the 16-foot bottom.
b2_s0	Set 1: Two steps: what is the total perimeter of all 3 panels? Feet:	**108** (also accepted: 108 feet, 108 foot, 108 ft)	One panel: 10 + 16 + 5 + 5 = 36 feet. Three panels: 3 × 36 = 108 feet.
b1_s1	Set 2: *Ximena and Ms. Whitaker Finish the Design The new truss is a row of equilateral triangles. Each triangle has a perimeter of 27 feet. The bottom beam is made of 5 triangle sides in a row.* How should the triangles be classified by their angles? [Acute, because every angle of an equilateral triangle is 60° / Right, because trusses need right angles / Obtuse, because the triangles point up and down / It is impossible to tell without a protractor]	**Acute, because every angle of an equilateral triangle is 60°**	All three angles of an equilateral triangle are 60°, and 60° is less than 90°, so every equilateral triangle is acute.
b2_s1	Set 2: Two steps: how long is the bottom beam? Feet:	**45** (also accepted: 45 feet, 45 foot, 45 ft)	Each side: 27 ÷ 3 = 9 feet. Five sides in a row: 5 × 9 = 45 feet.

ID	Question	Answer	Why (explain to student)
b1_s2	Set 3: *Ximena and Mr. Wojcik Finish the Design Mr. Wojcik cuts a square steel plate with 20-inch sides along BOTH diagonals, making 4 triangles that meet in the center. The diagonals of a square are equal, cross at a right angle, and cut each other in half. He will make 6 square plates in all.* What is the full name of each of the 4 small triangles? [Right isosceles triangles / Acute equilateral triangles / Obtuse isosceles triangles / Right scalene triangles]	**Right isosceles triangles**	The diagonals cross at a right angle, so each triangle has a right angle in the center. The diagonals cut each other in half, so the two sides that meet at the center are equal: right isosceles.
b2_s2	Set 3: Two steps: Mr. Wojcik sands the outside edges of all 6 square plates before he cuts them. How many inches of edge does he sand? Inches:	**480** (also accepted: 480 inches, 480 inch, 480 in)	One square: 4 × 20 = 80 inches of edge. Six squares: 6 × 80 = 480 inches.

Pre & Post sheet answer guide

Many items are open-ended. Grade for genuine understanding and complete sentences, not exact wording. Sheet: grade4-5/prepost/KCE_Math_PrePost_G4-5W24.pdf.

PRE-LESSON WARM-UP

#	Prompt	Model answer / grading note
1	Review: 3 X's at 3/8 inch and 2 X's at 3/4 inch	a) 3 × 3/8 = 9/8 and 2 × 3/4 = 6/4 = 12/8; 9/8 + 12/8 = 21/8 = 2 5/8 inches. b) 3/4 − 3/8 = 6/8 − 3/8 = 3/8 inch.
2	Draw a triangle with one right angle	Any triangle with one square corner, marked with a small square.
3	Sides and angles of a quadrilateral; name two	4 sides and 4 angles. Any two of: square, rectangle, rhombus, trapezoid, parallelogram, kite.
4	Lines that never cross	parallel lines
5	Shape with exactly one line of symmetry	Answers vary (for example, an isosceles triangle with only two equal sides, an isosceles trapezoid, a heart, or the letter A). The dashed line must split it into two matching halves, and there must be no second line.

POST-LESSON CHALLENGE

#	Prompt	Model answer / grading note
1	Classify by sides AND angles: a) 6, 6, 6 ft b) 90°, 30°, 60°, no equal sides c) 110°, 35°, 35°, two equal sides	a) equilateral and acute (every angle is 60°) b) right scalene c) obtuse isosceles
2	Isosceles brace: perimeter 31 feet, third side 7 feet	31 − 7 = 24 feet for the two equal sides; 24 ÷ 2 = 12 feet each. Check: 12 + 12 + 7 = 31.
3	Lines of symmetry: square, non-square rectangle, equilateral triangle, scalene triangle	a) 4 b) 2 c) 3 d) 0. The square drawing shows the vertical, horizontal and both diagonal lines; the rectangle shows only the vertical and horizontal lines through the middle.

#	Prompt	Model answer / grading note
4	Xander says every rectangle is a square	Model: Xander is wrong. A rectangle and a square both have four right angles, but a square must also have four equal sides. A rectangle that is 6 inches long and 2 inches wide has four right angles but is not a square. Every square is a rectangle, but not every rectangle is a square.

★ CHALLENGE

#	Prompt	Model answer / grading note
1	Always, sometimes or never; draw a right trapezoid	a) sometimes (only when all four sides are equal) b) always (a rhombus has 2 pairs of parallel sides) c) never (any two sides of a triangle meet). The right trapezoid has exactly one pair of parallel sides, marked with matching arrows, and two right angles where one side meets both parallel sides.